



XXVI

International Conference of the Society for Human Ecology

ECOLOGY, POLITICS, AND JUSTICE: WORKING TOWARDS THE PLURIVERSE

University of Mons | June 18–21, 2025 | Mons, Belgium

Poster

Title: Bridging Science and Practice for Sustainable Tomato Cultivation

List of Authors: Sanghyo Moon¹ & Cory Whitney¹

Affiliation:

1. Institute of Crop Science and Resource Conservation (INRES), University of Bonn

Presenter contact: Sanghyo Moon (s62smoon@uni-bonn.de)

Abstract:

For scientific research to support sustainable agricultural development, farmers who are the intended recipients of scientific innovations must be involved. Laboratory results, for example, should produce products usable for farmers in their socio-ecological and often resource-constrained conditions. The **HortiPrimed** project, funded by the German Federal Organic Farming Scheme (BÖL), focuses on enhancing the sustainability of tomato cultivation by investigating the efficacy of priming tomato seedlings to induce later resistance to stress caused by climate change. To support HortiPrimed in achieving impact on the ground, we apply **Holistic Decision Analysis** to assess the farm-level financial and environmental viability of laboratory molecular research results. We use a holistic approach to integrate scientific insights and lived experiences of tomato farmers as they adapt to changing pests, diseases, and financial considerations. We developed the interactive **Optimal Priming** application to help farmers evaluate the benefits of various tomato varieties and to offer feedback to laboratory partners about the potential efficacy of their research. At the heart of **Optimal Priming** is a probabilistic model combining information from the lab and farmers to compare conditions with and without priming and support farm-level management decisions. The decision model behind **Optimal Priming** is widely-applicable for implementation research, enhancing collaboration and trust between academics and providing actionable results for advancing innovation.

Keywords (max. 5): Decision analysis, interactive research, Sustainable agriculture